PRE-SEED INVESTMENT MEMORANDUM • SPECIALE INVEST DEEPTECH SYNDICATE

The Universal Bidirectional Sign Language Protocol

Sammy is a bidirectional sign language translation engine. Wait—before you think this is just another CSR or charity grant, it is not. It is high-margin enterprise software addressing a mandatory legal compliance crisis. We are building the universal standard for everywhere and anywhere, using airport departure gates as our commercial beachhead.

TARGET RAISE	VALUATION CAP	CORPORATE STRUCTURE	OPERATIONAL RUNWAY
$750,000 SAFE	$6,000,000 Post-Money	Delaware Parent + India R&D	48 Months (Indian DeepTech)

Prepared for: Vishesh Rajaram | Arjun Rao | Speciale Invest Investment Committee

Confidential Institutional Brief

FIRST-PRINCIPLES PROBLEM

Why Existing AI & Text Architectures Fail Sign Language

Sign Language is not spoken English with hand gestures; it is an independent 3D visual-spatial language with its own distinct grammar, phonology, and syntax.

1. The Text Captioning Illusion (Ava, Otter)

Linguistic Failure

70%+

Prelingually Deaf Read English Below 4th-Grade Level

Sign language is a spatial language, completely native and separate from spoken syntax. Subtitles and English transcripts fail during stressful, rapid airport disruptions or emergency announcements.

- Deaf travelers cannot parse dense, rapidly scrolling English text during flight gate swaps.
- Captions provide **zero bidirectional capability**: a deaf passenger cannot sign back to ask where their gate is.

2. The Generative Video Dead-End (Sora, Runway)

DeepTech Dead-End

$0.10 / sec

Unsustainable Cloud GPU Streaming Burn

Statistical pixel diffusion models hallucinate finger counts, cause severe skeletal drift ($\Delta > 0.12$ bone length error), and exhibit 15+ second latency.

- Fingers merge, clip through faces, and destroy the precise spatial grammar required for semantic comprehension.
- Streaming multi-million-dollar cloud GPU clusters onto public airport gate screens makes unit economics impossible.

Linguistic Citation: Traxler (2000), J. Deaf Studies & Deaf Education • NIH / Gallaudet Research

Linguistic & Technical Dilemma

THE DEEPTECH ARCHITECTURE

BiSign-LLM: The Deterministic Two-Way Spatial Engine

Replacing statistical pixel hallucination with rigid spatial constraint solvers executing client-side with **$0.00 cloud GPU cost**.

Hearing → Deaf (Forward)

3D Kinematic Avatar

Δ = 0.000

Rigid Bone Length Invariance

Airlines/hospital staff speak naturally; Sammy parses the audio, generates continuous spatial tokens, and compiles them via continuous quaternion SLERP over a 280ms physiological co-articulation window into fluid 3D avatars.

Deaf → Hearing (Backward)

Spatial Vision Translation

< 150 ms

Continuous Real-Time Latency

Deaf user signs into any standard commodity webcam. Sammy's **Causal Spatial Memory Bank (M_t)** tracks 3D spatial bone coordinates and non-manual facial cues, translating signs back into natural spoken voice and text.

Edge Execution ($0 Cloud GPU)

Pure Software Margin

60 FPS

In-Browser WebGL / Wasm Execution

Runs directly on existing client hardware—10-year-old Intel Atom airport display chips, bank teller monitors, or iOS/Android devices. Marginal compute cost per sign is $0.00, unlocking 90%+ gross margins.

Architecture: WebGL / Three.js / WebAssembly Procedural Engine (Zero Server GPU Inference)

Core Engine

ENGINEERING BENCHMARKS

Exponential over Incremental: Architectural Comparison

Why SAMMY is structurally defensible against Big Tech foundation models, text captioning tools, and legacy avatars.

ARCHITECTURAL DIMENSION	GENERATIVE VIDEO (SORA / RUNWAY)	TEXT CAPTIONS (AVA / OTTER)	LEGACY 3D (SIGNALL / DEAFTAWK)	SAMMY (BISIGN KINEMATIC ENGINE)
Biomechanical Bone Rigidity	**Hallucinates (Δ > 0.12)**	**Not Applicable (Text)**	**Robotic / Stiff Keyframes**	**Exact Invariance (Δ = 0.000)**
Inference Latency	15,000 ms – 30,000 ms	800 ms – 1,500 ms	1,200 ms	< 150 ms (Real-Time 60 FPS)
Cloud GPU Cost / Minute	$6.00 / minute streaming	$0.01 / minute	$0.08 / minute server-side	$0.00 (100% Client-Side Edge)
Two-Way Bidirectional Dialogue	**No (One-way video only)**	**No (English text only)**	**No (One-way avatar only)**	**YES: Speech-to-Sign & Sign-to-Speech**
Enterprise Hardware CapEx	Requires High-End Server GPU	Zero (Software)	Custom specialized cameras	Zero (Integrates on existing displays)
14 CFR § 382 Federal Compliance	Non-compliant (Hallucinations)	Partial (Fails low-literacy signers)	Partial (Uncanny valley rejection)	100% Fully Compliant

Benchmark: Evaluated against US DOT 14 CFR § 382.53 Federal Transit Display Requirements

Competitive Moat

COMMERCIAL BEACHHEAD

The High-Margin Beachhead: Airport Departure Gates

Driven by federal statutory compliance mandates, zero hardware CapEx, and finite operational vocabularies.

The Regulatory Forcing Function: 14 CFR § 382.53

Federal Mandate

$10,000 / Gate

Annual Enterprise Display SaaS License

Under the US Department of Transportation's **Air Carrier Access Act (14 CFR § 382.53)**, commercial airlines and airport operators face statutory six-figure federal consent decrees if dynamic flight gate changes and emergency announcements are not visually accessible to deaf passengers.

- **Zero Hardware CapEx:** Deploys directly onto existing gate monitors and flight information display systems (FIDS) via lightweight browser runtime.
- **Low Operational Liability:** Finite travel vocabulary (~250 operational phrases) with zero medical malpractice exposure.
- **Immediate High ARR:** A single hub like Dallas/Fort Worth (168 gates) or Atlanta (192 gates) represents $1.6M – $1.9M in annual recurring revenue.

The $20M+ Ava Footprint is Our Warm Pipeline

Proven Budget Lines

Ava proved that enterprise accessibility budgets already exist, selling text captioning to **Air France, UCLA, Autodesk, and government agencies**.

- Every enterprise paying Ava today has an **unresolved spatial sign language liability** for native Deaf signers who cannot read fast English captions.
- We do not need to educate procurement departments on accessibility budgets; we provide the spatial 3D signing engine that satisfies their legal mandate.
- Immediate expansion: Airport Gates → Rail Terminals (Amtrak) → Hospital Triage → Bank Teller Counters.

Federal Compliance: US Department of Transportation 14 CFR § 382.53(a)(3)

Go-To-Market Beachhead

BUSINESS MODEL & GROWTH FLYWHEEL

The Dual Engines: $10K/Gate SaaS + $2/Month Consumer Mobile

Combining high-margin enterprise compliance ARR with mass consumer network effects across 70M+ deaf individuals globally.

Engine 1: B2B & B2B2C Enterprise SaaS

Cash Flow Engine

90%+

Software Gross Margins (Client-Side Edge Compute)

High-ACV enterprise compliance software sold to airlines, airport authorities, hospitals, and transit hubs:

- **Annual License:** $10,000 per departure gate / public terminal annually.
- **B2B2C Dynamics:** Enterprise pays the bill; thousands of daily deaf travelers and hearing staff use the bidirectional interface.
- **High Switching Costs:** Once integrated into airport FIDS and ground crew operations, Sammy becomes standard operational infrastructure.

Engine 2: B2C Mobile Direct Subscriptions

Grassroots Moat

$2.00 / mo

Mass-Market Accessible Smartphone Pricing

Over 70 million deaf individuals globally need everyday communication for ordering coffee, visiting retail stores, and talking to hearing peers:

- **Zero CAC Flywheel:** Travelers experience Sammy on airport monitors and download the smartphone app for personal use.
- **Unassailable Data Moat:** Everyday consumer signing interactions continuously feed our causal spatial memory bank, expanding our real-world linguistic edge.
- **Community Trust:** Transparent, affordable pricing cements deep cultural loyalty within Deaf associations worldwide.

Unit Economics: Edge Execution eliminates server streaming bills, enabling both enterprise scale and $2/mo pricing

Business Model

LONG-TERM PLATFORM HORIZON

The Universal Standard: Sammy for Everywhere & Anywhere

Capturing airport gates first to establish regulatory cash flow, then scaling Sammy into the universal foundation protocol for deaf communication.

Healthcare & Emergency

Life-Critical

< 30 sec

Instant Emergency Room Triage

Hospitals currently wait 45+ minutes for live video-relay interpreters during medical emergencies. Sammy enables instantaneous two-way bedside diagnosis with full HIPAA compliance and zero data leaks.

Civic, Banking & Retail

Everyday Commerce

100,000+

Target Countertop Terminals

Bank teller stations, government DMV offices, and drive-thru kiosks integrate Sammy's WebAssembly widget, making physical counter service accessible to every citizen without hiring specialized staff.

Video Calls & Mobile OS

Universal Software Layer

SDK / API

Lightweight WebRTC Plug-in

Embedding Sammy directly into Zoom, Microsoft Teams, iOS, and Android. Any hearing participant talks normally, and any deaf attendee sees real-time signing on screen and signs back seamlessly.

Vision: Becoming the "TCP/IP" and Universal Runtime for Sign Language Worldwide

Platform Expansion

SPECIALE PORTFOLIO SYNERGY

Parallel DNA: How SAMMY Fits Speciale Invest

Mirroring the rigorous engineering discipline of Speciale Invest's most successful deeptech investments.

Like Morphing Machines

Compute Agility

Edge Efficiency over Server GPU Burn

Just as Morphing Machines rethought parallel compute architectures out of IISc, SAMMY rejects multi-million-dollar cloud GPU clusters to execute deterministic spatial kinematics directly on edge display chips.

Like CynLr (Robotics)

Physics Moats

Spatial Kinematics over Guesswork

Just as CynLr solves robotic vision through physical convergence rather than statistical ML guesswork, SAMMY enforces rigid skeletal mechanics and quaternion SLERP over hallucinating pixel diffusion.

Like Protecto & StreamAlive

Enterprise Rails

Low-Latency Infrastructure

Like Protecto's compliance layer for GenAI and StreamAlive's low-latency interactive display software, SAMMY builds the critical regulatory plumbing for enterprise public transit.

TEAM & CAPITAL EFFICIENCY

Founding Team & The 4-Year Indian Runway Arbitrage

S-Tier systems and spatial AI engineering operating at 1/5th Silicon Valley burn rate.

Founding Team & Engineering Complement

Co-Founders

2 Founders

Spatial AI + Systems / Compiler Engineering

Iniyan Joseph (Founder & Spatial AI Engineer): Research background in sign language kinematics, procedural 3D inverse solvers, and bidirectional translation architectures.

Co-Founder & CTO (Systems Architect): Expert C, C++, and Java background specializing in high-performance WebAssembly compilation, memory-constrained embedded runtimes, and native Android NDK integration.

- Founder & Co-founder are aligned under a 4-year vesting schedule with a 1-year cliff.
- Uncompromising focus on low-level systems engineering rather than brittle API wrappers.

The 48-Month Financial Sustenance Model ($750K / ₹6.24 Cr)

Unmatched Efficiency

CATEGORY	MONTHLY (INR)	48-MONTH TOTAL (INR)
Founders & Systems Engineers	₹5,50,000 / mo	₹2,64,00,000 ($316K)
Office / Engineering Hub	₹1,00,000 / mo	₹48,00,000 ($58K)
In-House MoCap Lab (One-Time)	—	₹25,00,000 ($30K)
US Corporate & International Patents	—	₹35,00,000 ($42K)
Airport Pilots & Compliance Testing	₹1,50,000 / mo	₹72,00,000 ($86K)
Strategic Reserve / Buffer	—	**₹1,80,00,000 ($218K)**

Unit Economics: 48 Months of runway allows out-surviving and out-executing any US competitor

Runway Economics

THE INVESTMENT OPPORTUNITY

The Ask: Inviting Speciale Invest to Lead Our Pre-Seed

A category-defining DeepTech investment at the intersection of spatial AI, public transit infrastructure, and civil rights.

The Offering Structure

Deal Terms

- **Target Raise:** $750,000 (Directly in Speciale Invest's pre-seed lead sweet spot).
- **Instrument:** Standard Y Combinator Post-Money SAFE note.
- **Valuation Cap:** $6,000,000 Post-Money.
- **Position:** Inviting Speciale Invest to serve as the **Lead Institutional Investor**.
- **Corporate Vehicle:** Delaware C-Corp parent holding all IP, with Indian wholly-owned operational R&D subsidiary.

Next Steps & Immediate Diligence Access

Live Proof

We did not wait for funding to build the engine. Our live 3D procedural studio is operational right now for independent diligence:

- **Interactive 3D Studio Demo:** Launch 3D WebGL Engine
- **Founder Profile:** linkedin.com/in/iniyandrews
- **Official Email:** iniyandrews@gmail.com
- **Technical Readiness:** Ready for immediate partner-level technical diligence with Speciale's investment team.

SAMMY • Ready for Immediate Diligence

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